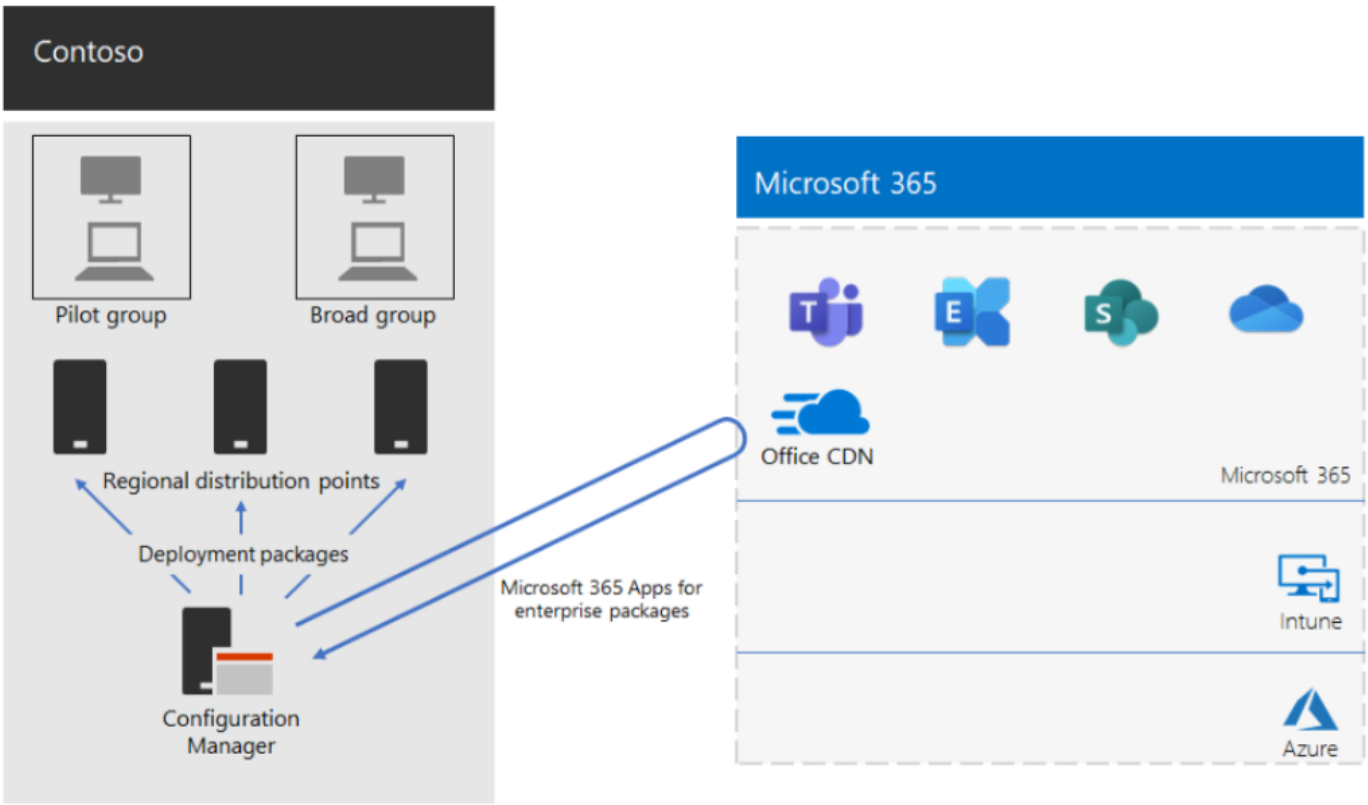
# Data Centers, Service Providers, and Cloud Computing Elements

Here is the Contoso Microsoft 365 Apps for enterprise installation and ongoing updates deployment architecture.




## A. Data Centers Certification Tiers (Uptime

- **Tier 1.**
- **Tier 2.**
- **Tier 3.**
- **Tier 4.**

## B. Data Centers Types/Modes of Operations

- Internet Exchange.
- Cloud Node.
- Colocation Facility.
- Disaster Recovery and Business Continuity (DRBC).

## C. Cloud Computing Elements

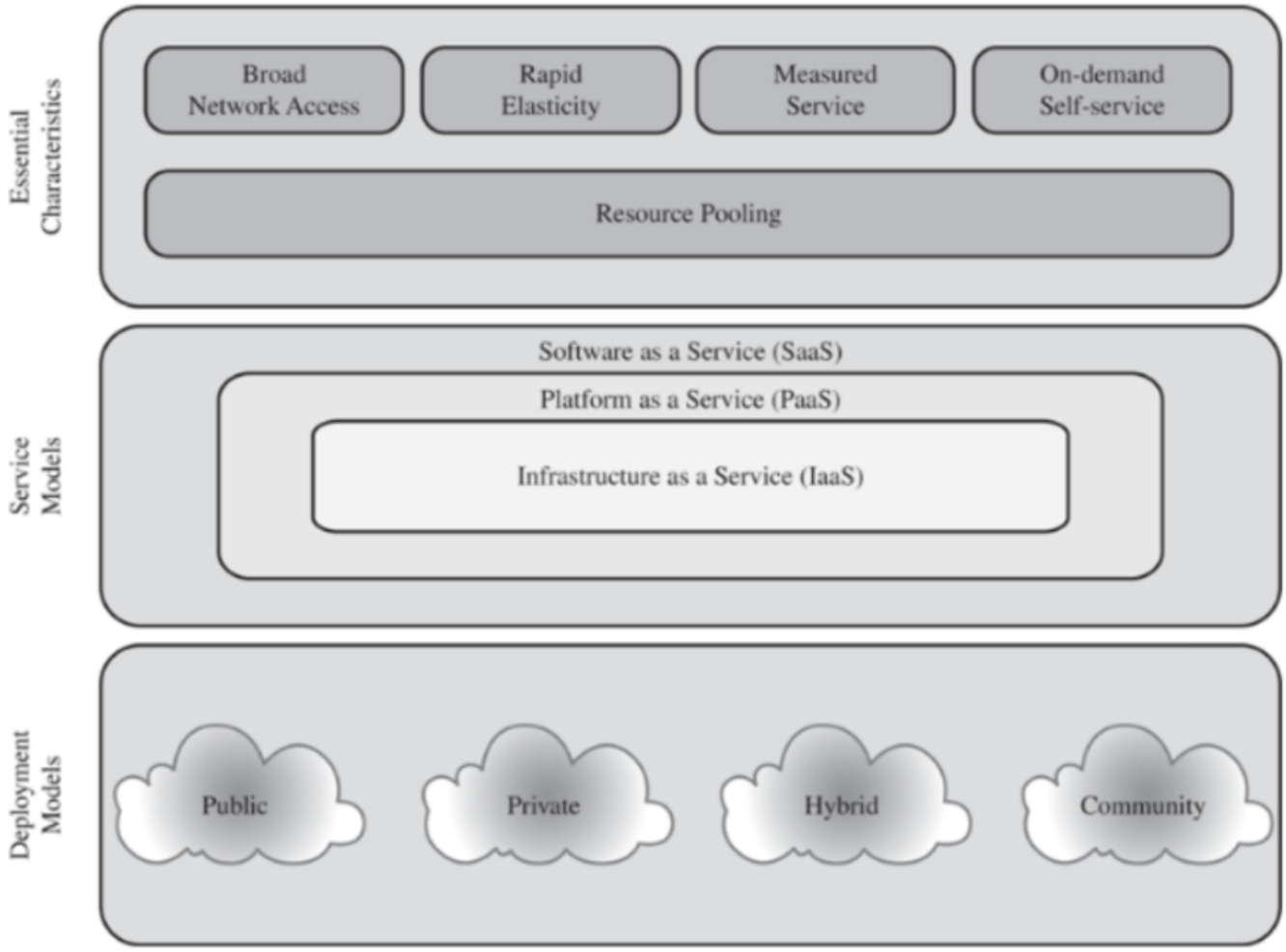



**Figure 13.1 Cloud Computing Elements**

**Table 13.1 Comparison of Cloud Deployment Models**

| | Private | Community | Public | Hybrid |
|---|---|---|---|---|
| **Scalability** | Limited | Limited | Very high | Very high |
| **Security** | Most secure option | Very secure | Moderately secure | Very secure |
| **Performance** | Very good | Very good | Low to medium | Good |
| **Reliability** | Very high | Very high | Medium | Medium to high |
| **Cost** | High | Medium | Low | Medium |

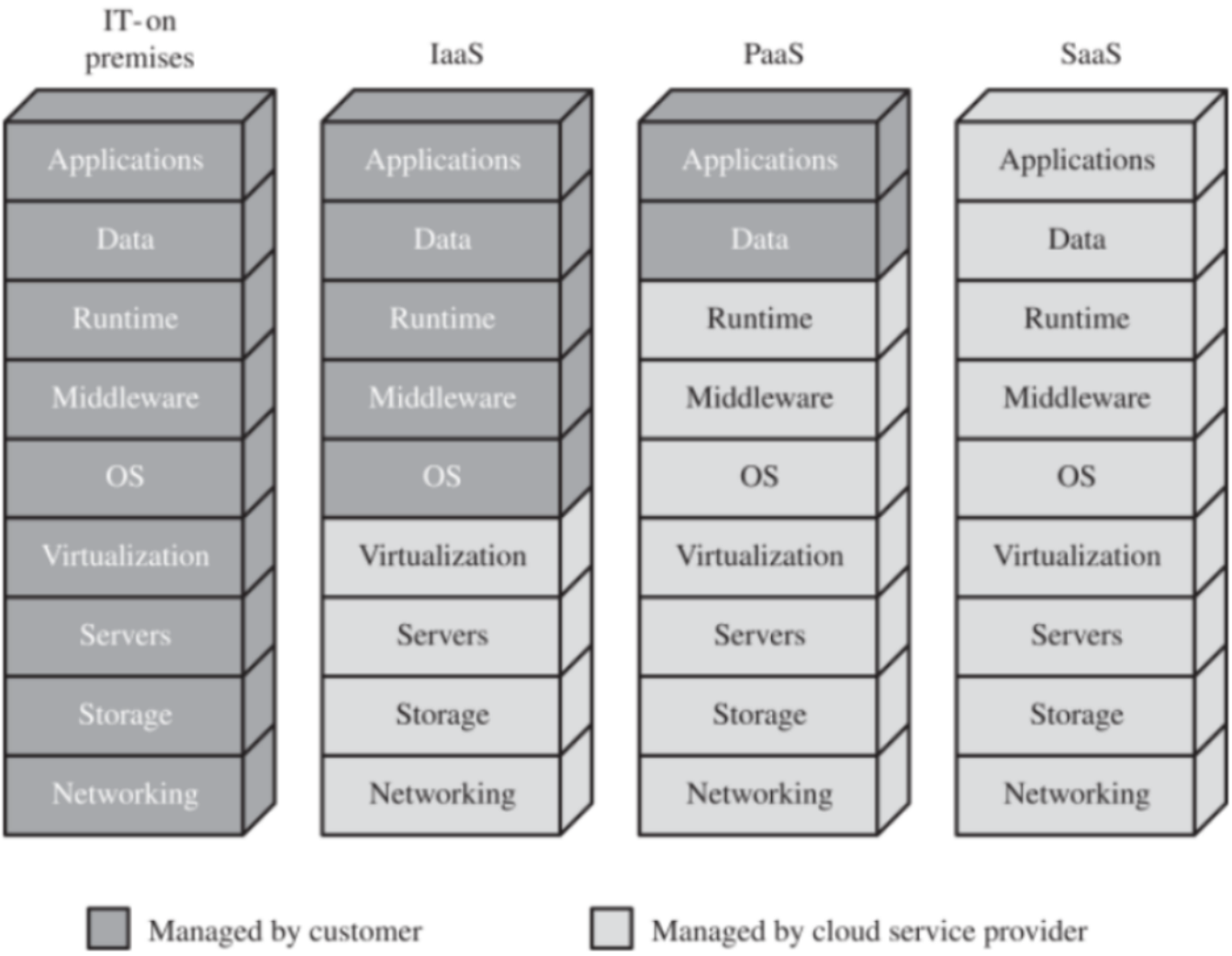
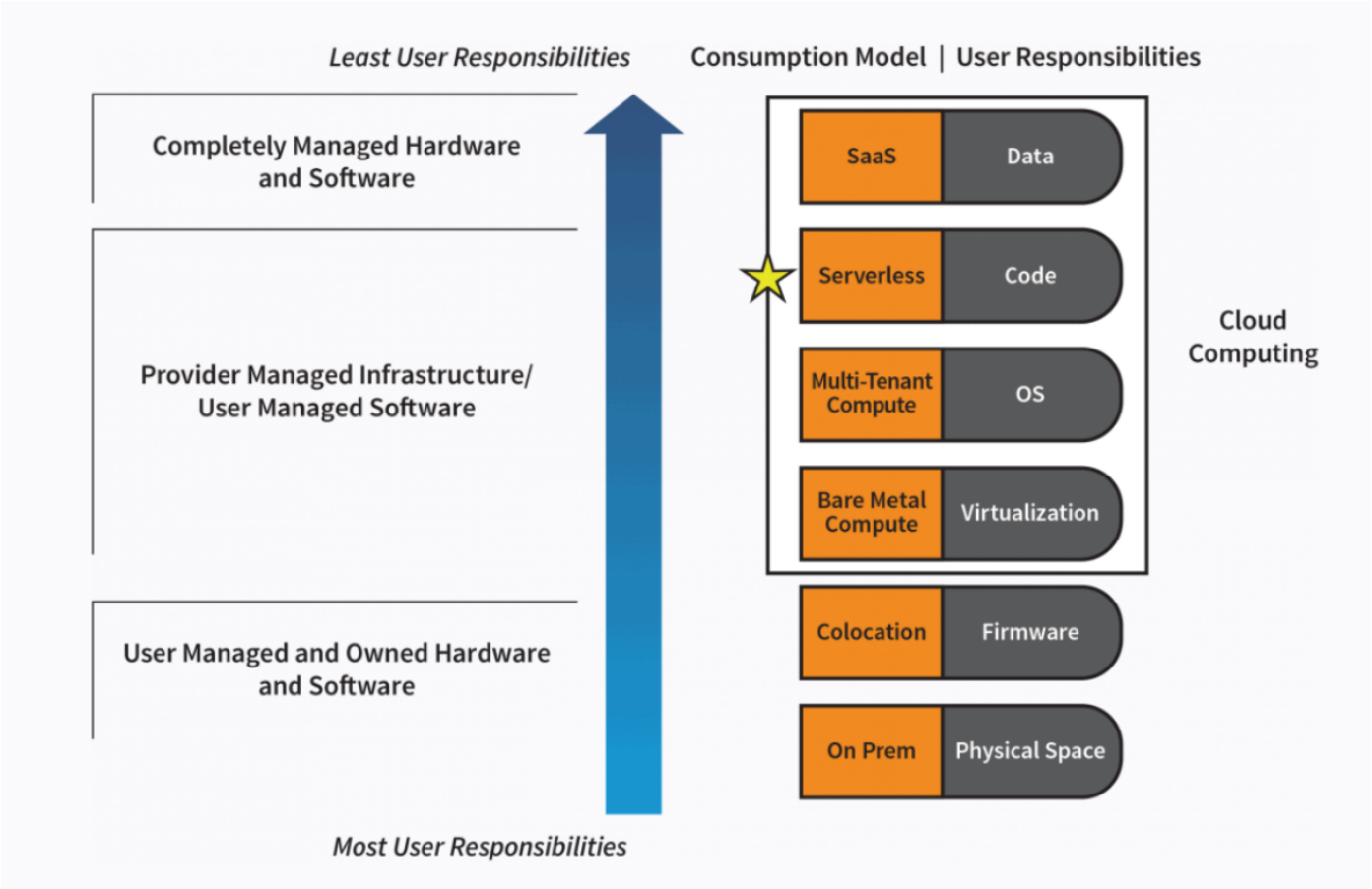



**Figure 13.2 Separation of Responsibilities in Cloud Service Models**



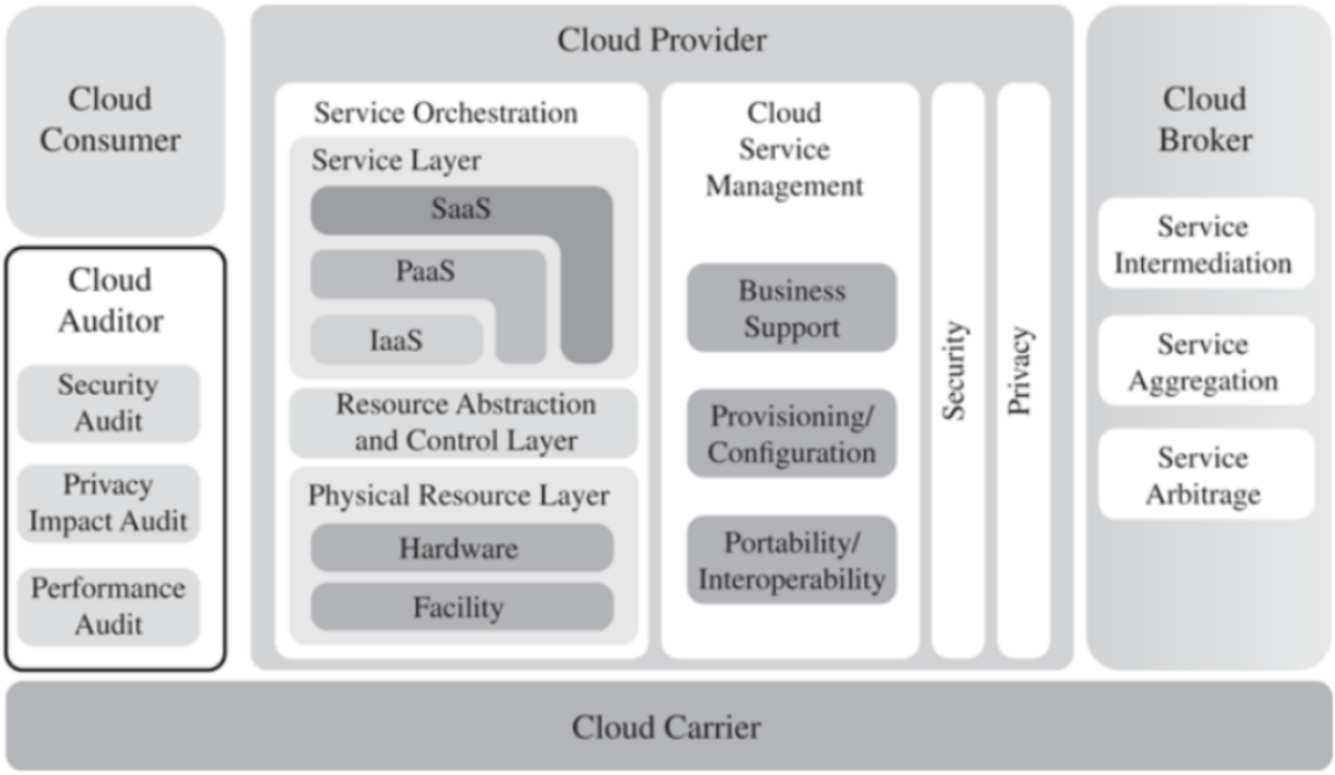




**Figure 13.3 NIST Cloud Computing Reference Architecture**

**Bing Videos (https://www.bing.com/videos/riverview/relatedvideo?&q=alibaba+data+center&&mid=FF0158B455518D0F9314FF0158B455518D0F9314&&FORM=VRDGAR)**